

Comparative analysis of sparse gene-regulatory network inference on murine gastruloid single-cell RNA-seq

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Abstract

Organoid and gastruloid models reduce the need for in vivo experiments, but interpreting their transcriptomes requires maps of gene–gene dependence. We outline a benchmark of gene-regulatory network (GRN) inference on gastruloid single-cell RNA-seq, restricted here to two sparse Gaussian graphical model (GGM) estimators—the frequentist package SILGGM and the Bayesian latent mixture BLGGM—and an optional directed tree ensemble (GRNBoost2) as a topological comparator. Networks are compared by engineering cost, runtime and edge-set overlap (raw and chance-adjusted Jaccard), not by a gold standard. The companion code lives in the DeCovarT reproducibility repository.

Keywords: gene regulatory network, Gaussian graphical model, scRNA-seq, gastruloid

1 Introduction

To study genetic disease while reducing in vivo work, biologists grow three-dimensional structures that mimic organ development (organoids). Gastruloids are organoids that recapitulate gastrulation-stage patterning. Understanding how genes jointly build tissue requires GRN inference: a weighted graph or an adjacency table (Fig. 1).

This manuscript recasts an M1 internship benchmark on gastruloid scRNA-seq from Suppinger and colleagues [1]. We drop graph-neural methods that inject literature embeddings and keep two precision-matrix estimators: SILGGM [2] and BLGGM [3], both descendants of the graphical lasso [4]. GRNBoost2 remains a directed, dense tree-ensemble comparator [5], as used in SCENIC [6].

2 Methods

2.1 Data

We use gastruloid scRNA-seq [1]. Rows are cells and columns are genes. The original object has 2,819 cells and about 27,000 genes. We retain the 500 genes of largest marginal variance as a computational filter, not as a leakage-free signature for deconvolution.

2.2 Hardware

Experiments were run on the laboratory cluster: Ubuntu 22.04.5 LTS, 40 CPU cores, NVIDIA RTX 5000 Ada (32 GB), CUDA 12.2. Reproducible R jobs in this repository use at most two cores on CI.

2.3 Estimators

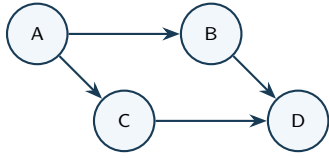
All methods take the same filtered expression table (plus optional transcription-factor lists for GRNBoost2).

GRNBoost2.

Gradient-boosted trees per target gene in the Arboreto framework [5], the scalable engine used by SCENIC [6]. Input: expression matrix and a transcription-factor dictionary. Output: a directed, typically dense network.

BLGGM.

Bayesian latent Gaussian graph mixture [3]. Latent Gaussians per cell type, non-ignorable dropout, spike-and-slab precision matrices. Output: sparse undirected graphs after MCMC.



Graph (edge = interaction)

	A	B	C	D
A	0	1	1	0
B	1	0	0	1
C	1	0	0	1
D	0	1	1	0

Fig. 1 A GRN as an undirected graph (left) and a binary adjacency table (right). Entry 1 marks an edge; 0 marks none. Layout follows the internship schematic without raster artwork.

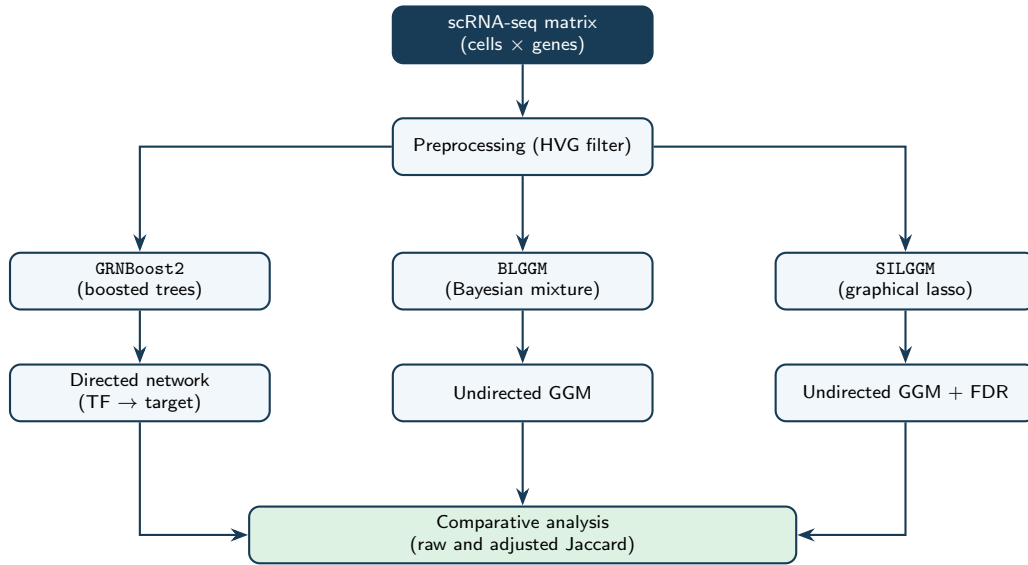


Fig. 2 GRN benchmark pipeline after dropping literature-embedding graph networks. GRNBoost2, BLGGM and SILGGM share the same filtered matrix; Jaccard is computed on symmetrised edge sets.

SILGGM.

Frequentist high-dimensional GGM inference with FDR control [2]. After a nonparamormal transform [7] implemented in [huge](#) [8], de-sparsified graphical lasso or related tests yield a sparse undirected graph.

2.4 Benchmark pipeline

GRN inference is unsupervised: there is no tissue-wide gold standard. We therefore compare consensus among methods (Fig. 2). Directed edges from GRNBoost2 are symmetrised before Jaccard comparison with undirected GGMs.

Table 1 Software characteristics (internship assessment).

Method	Stack	Install
GRNBoost2	Python / Arboreto	Easy
SILGGM	R / CRAN	Medium
BLGGM	R / GitHub, OpenMP	Hard

3 Results

We score engineering feasibility, computation, and topological overlap. Reported numbers are those of the internship on the 500-gene gastruloid slice; they are not re-estimated in this template.

3.1 Engineering

Table 1 summarises installation burden. Easy means a standard language stack; Medium means CRAN/R configuration; Hard means compiling native code (OpenMP) or pinning GPU stacks.

3.2 Computation

On the 500-gene table, GRNBoost2 finished in about 30 s and returned 26,615 directed edges. BLGGM needed about nine hours of MCMC for 3,029 high-confidence undirected edges. SILGGM sits between these extremes in the frequentist GGM class; exact wall times belong in the companion scripts once inputs are frozen.

3.3 Overlap

Let A and B be edge sets. Raw Jaccard is

$$J_{\text{raw}}(A, B) = \frac{|A \cap B|}{|A \cup B|}. \quad (1)$$

Coverage differs sharply across methods, so we also use a chance-adjusted index

$$J_{\text{adj}} = \frac{J_{\text{raw}} - E[J_{\text{raw}}]}{1 - E[J_{\text{raw}}]}. \quad (2)$$

BLGGM and SILGGM, despite Bayesian versus frequentist implementations, showed a stronger independent consensus on partial-correlation edges than either did with the dense GRNBoost2 graph. That pattern is expected if both GGMs filter confounding co-expression.

4 Discussion

Limitations include the lack of a gold-standard network, sensitivity to regularisation (lasso λ , FDR α , MCMC length, mixture K), and runtime disparity. Nonparanormal

shrinkage or truncation [7] in **huge** [8] is the default Gaussian-copula pre-step for SILGGM because raw counts are not Gaussian. BLGGM instead models dropout on library-normalised non-negative data [3].

Code and launch headers for **Rscript/**`nohup` are in https://github.com/bastienchassagnol/DeCovarT_reproducibility.

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